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Monsoon, 2025
B. Tech 3rd Semester Examination
Mathematics To Civil Engineering
Course Code: BMA23211T

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

Answer the following questions

1. Choose the correct options:

1x10=10

- a) The period of the function $f(x) = \cos 2x$ is
 - (i) $\frac{\pi}{2}$
 - (ii) π
 - (iii) 2π
 - (iv) $\frac{2\pi}{3}$
- b) The function $f(x) = x \sin x, -\pi < x < \pi$ is
 - (i) an odd function
 - (ii) an even function
 - (iii) neither an odd nor an even function
 - (iv) both an odd and an even function
- c) The Fourier series of even function contains only
 - (i) sine terms
 - (ii) cosine terms
 - (iii) both sine and cosine terms
 - (iv) none of these
- d) $L\{t^5\} =$
 - (i) $\frac{5}{s^5}$
 - (ii) $\frac{5}{s^6}$
 - (iii) $\frac{5!}{s^5}$
 - (iv) $\frac{5!}{s^6}$
- e) If $L\{f(t)\} = \bar{f}(s)$, then $L\{tf(t)\} =$
 - (i) $\frac{\bar{f}(s)}{s}$
 - (ii) $s\bar{f}(s)$

- (iii) $\frac{d}{ds} \bar{f}(s)$
 (iv) $-\frac{d}{ds} \bar{f}(s)$
- f) $L^{-1}\left\{\frac{1}{s^5}\right\} =$
 (i) $\frac{t^4}{4!}$
 (ii) $\frac{t^5}{4!}$
 (iii) $\frac{t^4}{5!}$
 (iv) $\frac{t^4}{5!}$
- g) $L^{-1}\left\{\frac{1}{s^2-9}\right\} =$
 (i) $\frac{1}{3} \sin 3t$
 (ii) $\cos 3t$
 (iii) $\frac{1}{3} \sinh 3t$
 (iv) $\cosh 3t$
- h) The degree of the partial differential equation $\frac{\partial^3 z}{\partial y^3} = \left(\frac{\partial z}{\partial x} + z\right)^{\frac{3}{2}}$ is
 (i) 1
 (ii) 2
 (iii) 3
 (iv) 1.5
- i) The one-dimensional Wave equation is given by
 (i) $\frac{\partial y}{\partial x} = c^2 \frac{\partial^2 y}{\partial t^2}$
 (ii) $\frac{\partial y}{\partial t} = c^2 \frac{\partial^2 y}{\partial x^2}$
 (iii) $\frac{\partial^2 y}{\partial x^2} = c^2 \frac{\partial^2 y}{\partial t^2}$
 (iv) $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$
- j) If $\sin \frac{2\pi x}{a} = a_1 \sin \frac{\pi x}{a} + a_2 \sin \frac{2\pi x}{a}$, then
 (i) $a_1 = a, a_2 = 1$
 (ii) $a_1 = 1, a_2 = 1$
 (iii) $a_1 = 0, a_2 = 1$
 (iv) $a_1 = 1, a_2 = 0$

2. Answer **any one**:

a) Expand the function in Fourier Series $f(x) = x + x^2, -\pi \leq x \leq \pi$ 10

b) Find a Fourier Series of the function $f(x) = \begin{cases} 0, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$ 10

c) Find a Fourier cosine series of $f(x) = x, 0 < x < 2$. Hence deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$
8+2=10

d) Find the Fourier series expansion of the function $f(x) = \begin{cases} 0, -5 < x < 0 \\ 3, 0 < x < 5 \end{cases}$ 10

3. Answer any two:

a) Evaluate: (i) $L\{\sqrt{t}\}$, (ii) $L\{\sin^2 4t\}$, (iii) $L\{t \cos 2t\}$ 2+4+4=10

b) Evaluate: (i) $L\{e^{2t}(2\cos 3t + \sin 3t)\}$, (ii) $L\left\{\frac{1-\cos 2t}{t}\right\}$ 4+6=10

c) Evaluate: (i) $L^{-1}\left\{\frac{s+2}{s^2-5s+6}\right\}$, (ii) $L^{-1}\left\{\frac{2s-3}{s^2+4s+2}\right\}$ 4+6=10

d) Evaluate: (i) $L^{-1}\left\{\frac{1}{s^3(s^2+1)}\right\}$

(ii) $L^{-1}\left\{\frac{1}{(s+2)(s+3)}\right\}$ (using Convolution Theorem) 4+6=10

e) Solve the differential equation $x'' + 3x' - 3x = \sin t$, given that $x(0) = x'(0) = 0$ 10

4. a) A tightly stretched string of length l is initially in a position given by $y = a \sin \frac{\pi x}{l}$. If it is released from rest from this position, find the displacement $y(x, t)$. 10

OR

b) Solve the one-dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ subject to boundary conditions

$u(0, x) = 0 = u(l, t)$ and initial condition $u(x, 0) = x$
